ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE B

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Capacitors, relays and waveforms; motor control from two places; hydraulic symbols, the power pack and meter-in speed control; electro-hydraulic automatic return; and maintenance and fault-finding.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Draw all circuits and symbols neatly, using a ruler and standard symbols.
5. Use only a black or blue pen.
6. Write neatly and legibly.

QUESTION 1

1.1 Define the term *capacitor*. (2)

1.2 State TWO uses of capacitors in electronic circuits. (2)

1.3 Differentiate between a *relay* and a *contactor*. (2 × 2) (4)

1.4 FIGURE 1 below shows the output of a square-wave signal generator.

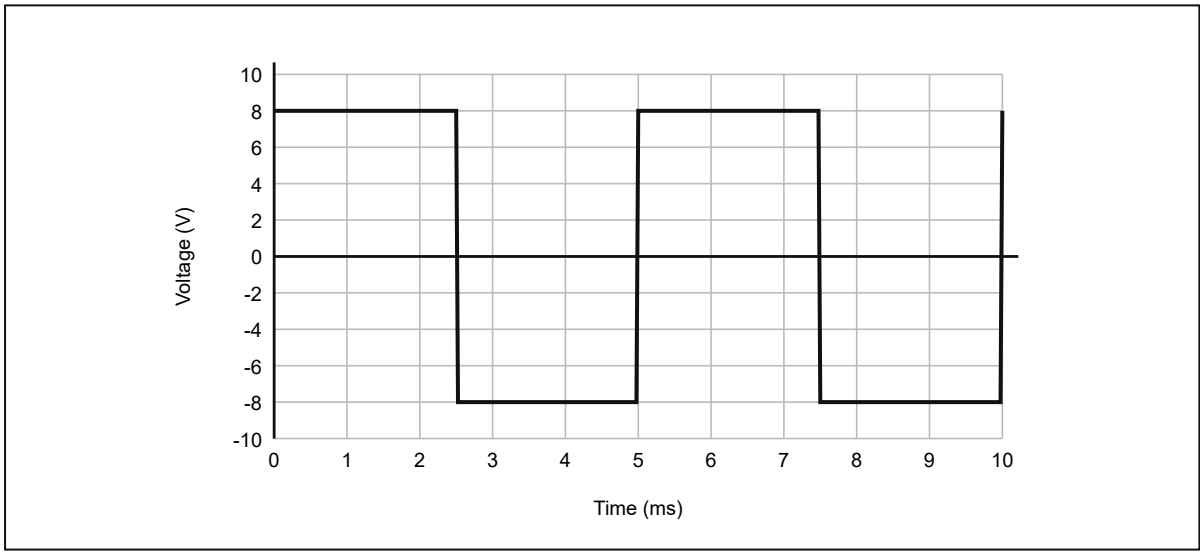


FIGURE 1

Use FIGURE 1 to determine the following:

1.4.1 The amplitude of the signal (1)

1.4.2 The peak-to-peak voltage (1)

1.4.3 The period of the signal (1)

1.4.4 The frequency of the signal (show ALL calculations) (3)

1.5 Draw a fully labelled symbol of a Zener diode and state ONE use of a Zener diode. (6)

[20]

QUESTION 2

2.1 Explain the function of an overload relay in a motor control circuit. (2)

2.2 Name TWO types of AC motors used in industry. (2)

2.3 Give ONE reason why a stepper motor is used instead of a DC motor to position a robot arm. (2)

2.4 Explain why a holding (latching) contact is used in a start/stop circuit. (2)

2.5 Design a relay control circuit in which a motor can be started from TWO places (START 1 and START 2) and stopped from TWO places (STOP 1 and STOP 2). The circuit must have a holding contact and an overload contact (F1). (12)

[20]

QUESTION 3

3.1 Draw the symbol of EACH of the following:

3.1.1 One-way flow control valve (3)

3.1.2 Single-acting cylinder with spring return (2)

3.2 Label the parts of the hydraulic power pack shown in FIGURE 2 below. Write only the answer next to the question number (3.2.1–3.2.6) in the ANSWER BOOK.

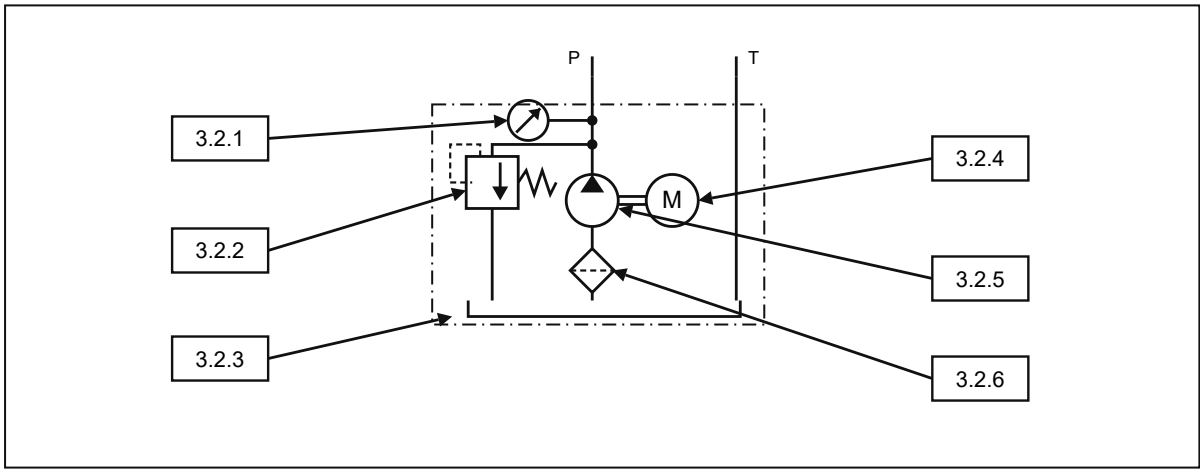


FIGURE 2

(6 × 1) (6)

3.3 Design a hydraulic circuit that controls the speed of a double-acting cylinder while it extends, using meter-in control. Make use of: a 4/2-way valve with a single solenoid (Y1) and spring return; a one-way flow control valve; and a hydraulic power pack. (9)

[20]

QUESTION 4

4.1 Explain the function of a pressure switch in an electro-hydraulic system. (2)

4.2 Name TWO types of sensors that can detect the end position of a cylinder. (2)

4.3 A double-acting cylinder is controlled by a 4/2-way double-solenoid valve (Y1 extends, Y2 retracts). Design the electro-hydraulic working circuit and the electrical control circuit so that: pressing S1 makes the cylinder extend (relay K1 holds itself); when the cylinder reaches the end of its stroke, limit switch S3 releases K1 and switches on Y2 so the cylinder retracts automatically; lamp H1 is on while the cylinder extends; and an emergency stop removes the power from the control circuit. (16)

[20]

QUESTION 5

5.1 Explain the term *corrective maintenance*. (2)

5.2 State FOUR checks you would carry out before starting a hydraulic system. (4)

5.3 A DC motor in a relay control circuit does not start when the START button is pressed. Give FOUR possible causes. (4)

5.4 Explain how an insulation resistance tester is used to test the windings of a motor. (5)

5.5 Explain why hydraulic oil must be kept clean, and give TWO ways of keeping it clean. (5)

[20]

TOTAL: 100

END OF QUESTION PAPER